IGCSE 0607 Practice Questions — E1.7 — Combined Questions & Markscheme

Indices I

Total Questions: 3 — **Total Marks:** 17 Marks

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  → **Paper 2 (Non-Calculator)** *(3 Qs)*

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1 **(a)** Work out the value of

$125^{\frac{1}{3}}$

..... **[1]**

(b) Work out the value of

6^{-2}

..... **[1]**

(c) Simplify.

$t^3 \times t^5$

..... **[1]**

2 **(a)** Find the exact value of

$16^{-\frac{3}{4}}$

..... **[2]**

(b) Simplify fully.

$(5x^3y^{-2})^2$

..... **[2]**

(c) Solve the equation.

$2^{3x-1} \times 4^x = 32$

$x =$ ..... **[3]**

3 **(a)** Simplify fully.

$$\left(\frac{16x^8}{y^4}\right)^{-\frac{1}{2}}$$

..... **[3]**

(b) Solve the equation.

$$4^x - 2^{2x+1} + 8 = 0$$

$x =$ ..... **[4]**

1 **(a)** 5 **[B1]**

(b) $\dfrac{1}{36}$ **[B1]**

(c) t^8 **[B1]**

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2 **(a)** $16^{\frac{1}{4}} = 2$ or equivalent correct first step **[M1]**

$\dfrac{1}{8}$ **[A1]**

(b) 25 **[B1]**

x^6y^{-4} or $\dfrac{x^6}{y^4}$ **[B1]**

(c) Expresses terms with a common base, e.g. $4^x = 2^{2x}$ or $32 = 2^5$ **[M1]**

Sets up a correct linear equation from indices, e.g. $3x - 1 + 2x = 5$ **[M1]**

$x = 1.2$ or $x = \dfrac{6}{5}$ **[A1]**

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3 **(a)** Inverts fraction to remove negative power or correctly applies negative power to all terms, e.g. $\left(\dfrac{y^4}{16x^8}\right)^{\frac{1}{2}}$ or $\dfrac{16^{-\frac{1}{2}}x^{-4}}{y^{-2}}$ **[M1]**

Correctly evaluates the coefficient $16^{-\frac{1}{2}} = \dfrac{1}{4}$ or correctly halves powers **[M1]**

$\dfrac{y^2}{4x^4}$ or $0.25x^{-4}y^2$ **[A1]**

(b) Recognises that $4^x = 2^{2x}$ or substitutes $u = 4^x$ **[M1]**

Expresses 2^{2x+1} as 2×2^{2x} or $2(4^x)$ **[M1]**

Simplifies to a single exponential term, e.g. $-2^{2x} = -8$ or $u - 2u = -8$ **[M1]**

$x = 1.5$ or $x = \dfrac{3}{2}$ **[A1]**

[Back to Question] *[Total Marks: 7]*